

Whole-Body Control for Autonomous FAST Ultrasound Scanning on a Legged Manipulator

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Abstract—Focused Assessment with Sonography in Trauma (FAST) is a rapid bedside ultrasound protocol that is critical in emergency medicine for detecting life-threatening internal injuries. Performing FAST scans autonomously with a mobile robot could extend emergency care to mass-casualty events, disaster sites, and scenarios where trained sonographers are unavailable. However, autonomous FAST scanning on a mobile platform poses two fundamental challenges: the robot must navigate to an unknown patient location using uncertain visual detection, and must simultaneously position the ultrasound probe with millimetre-level accuracy over the target anatomy.

This paper presents an integrated system for autonomous FAST scanning mounted on a Boston Dynamics Spot legged robot equipped with a Unitree Z1 manipulator arm. We formulate Whole-Body Control (WBC) as a single Quadratic Program (QP) that simultaneously drives the non-holonomic Spot base and the six-Degree of Freedom (DOF) Z1 arm, avoiding the sequential stop-then-manipulate paradigm common in existing mobile manipulation literature. The holistic Jacobian couples base and arm velocities through a manipulability-weighted damped least-squares formulation, automatically redistributing effort from the arm to the base near singular configurations. Patient localisation relies on a dual-camera pipeline: an Orbbec RGB-D (RGBD) camera provides wide-field You Only Look Once (YOLO)-based skeleton detection with Kalman Filter (KF)-filtered 3D torso tracking during navigation, while an Intel RealSense camera mounted at the End-Effector (EE) takes over for fine-grained surface estimation during scanning. A six-state Finite State Machine (FSM) orchestrates the mission phases—approach, sensor handoff, scanning, and reactive workspace extension—ensuring safe and reliable operation across a variety of patient positions and orientations.

Experiments on a physical Spot+Z1 platform demonstrate that the proposed WBC reduces mission completion time by **XX %** compared to a stop-and-manipulate baseline, while maintaining a probe positioning error below **XX mm**. To the best of our knowledge, this is the first system to demonstrate fully autonomous FAST scanning on a legged mobile manipulator.

Index Terms—Whole-body control, legged manipulation, medical robotics, FAST ultrasound, mobile manipulation, Kalman filtering, quadratic programming.

I. INTRODUCTION

FOCUSED Focused Assessment with Sonography in Trauma (FAST) is a standardised bedside ultrasound protocol that enables clinicians to rapidly detect life-threatening internal injuries—haemopericardium, haemothorax, and free intraperitoneal fluid—at the point of care [1]. Its speed and non-invasiveness make it indispensable in trauma bays, pre-hospital emergency medicine, and military field operations. However, FAST requires a trained sonographer who is physically present and manually positions the probe at four canonical anatomical windows: the subxiphoid, right upper quadrant,

left upper quadrant, and suprapubic regions. In mass-casualty incidents, pandemics, or remote deployment scenarios, qualified personnel are often unavailable or overwhelmed, creating a critical bottleneck in triage.

Autonomous robotic ultrasound has the potential to address this gap. A comprehensive review of robot-assisted ultrasound [2] and recent fixed-base systems [3]–[5] demonstrate that high-quality probe positioning can be achieved without direct human guidance. In our own prior work [6], we showed that a collaborative robot arm on a fixed base can autonomously screen the abdominal aorta using impedance control and deep learning segmentation, achieving clinically relevant diameter measurements. However, all these systems share a fundamental limitation: they require the patient to be pre-positioned within the arm's reachable workspace, confining autonomous ultrasound to controlled clinical settings.

Legged platforms such as the Boston Dynamics Spot offer a qualitative step beyond this limitation. They can navigate stairs, rubble, and confined spaces inaccessible to wheeled robots, and can therefore reach patients wherever they lie. Mounting a serial manipulator arm on a legged base introduces, however, a fundamental control challenge: the arm workspace is small relative to the environment, so the base must actively cooperate with the arm throughout the mission. Simply stopping the base while the arm operates—the prevalent approach in mobile manipulation [7]—wastes time and fails when the target falls outside the reachable workspace. WBC formulations that simultaneously optimise base and arm motion [8], [9] address this, but have not yet been applied to perception-driven medical scanning tasks where the patient location is unknown and must be estimated on-the-fly from noisy visual detections.

This paper presents Trustworthy Emergency Robot for Efficient Support and Assistance (TERESA), a complete system for autonomous FAST scanning on a Boston Dynamics Spot legged robot equipped with a Unitree Z1 manipulator arm. Building on the fixed-base scanning competence established in [6], we extend autonomous ultrasound to a fully mobile platform capable of operating in unstructured environments. The core of the system is a whole-body controller that formulates base and arm motion as a single QP over a holistic 6×8 Jacobian, avoiding the stop-then-manipulate paradigm. A dual-camera perception pipeline—wide-field Orbbec RGBD for navigation and EE-mounted Intel RealSense for surface estimation—feeds a six-state FSM that orchestrates the full mission lifecycle from patient detection to completed scan.

The main contributions of this paper are:

- 1) A **WBC formulation** for a non-holonomic legged base coupled with a six-DOF arm, with manipulability-

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weighted effort redistribution between base and arm (Section IV).

- 2) A **dual-camera perception pipeline** that provides robust patient localisation across the full approach distance, with a principled sensor handoff from wide-field skeleton tracking to EE surface estimation (Section V).
- 3) A **six-state mission FSM** that integrates WBC, perception, impedance contact control, and reactive workspace extension into a unified autonomous scanning system (Section V).
- 4) **Experimental validation** on a physical Spot+Z1 platform across a 3×3 grid of patient positions and orientations, demonstrating a **XX %** reduction in mission time and probe positioning error below **XX mm** compared to a stop-and-manipulate baseline (Section VII).

To the best of our knowledge, this is the first system to demonstrate fully autonomous FAST scanning on a legged mobile manipulator operating in unstructured environments.

The remainder of this paper is organised as follows. Section II reviews related work in robotic ultrasound, mobile manipulation, and legged robot control. Section III formally defines the system and the problem. Section IV derives the WBC formulation. Section V describes the complete system architecture. Section VI details the experimental setup and Section VII presents the results and discussion. Section VIII concludes the paper.

II. RELATED WORK

A. Robotic Ultrasound Systems

Robot-assisted ultrasound has been an active research area for over two decades, driven by the need to reduce operator variability, enable remote diagnostics, and automate repetitive scanning procedures. Early systems focused on teleoperation, using robotic arms to replicate the hand movements of a remote sonographer [4]. More recent work has moved toward increasing levels of autonomy, with robots capable of autonomously positioning and maintaining contact of the probe on the patient's body.

A fundamental challenge in autonomous ultrasound is probe-skin contact control. Gilbertson and Anthony [10] proposed a force and position control system for freehand ultrasound that decouples normal contact force from lateral probe positioning, establishing a framework widely adopted by subsequent systems. Building on this, Dyck et al. [11] extended impedance control to arbitrary curved surfaces using discrete differential geometry, enabling consistent probe orientation on non-planar anatomy—a capability directly relevant to abdominal scanning.

Autonomous scan planning has been tackled from multiple angles. Suligoj et al. [12] presented RobUSt, a system that autonomously acquires medical images using a pre-defined anatomical protocol, while Duan et al. [5] demonstrated optimisation-based control for scoliosis assessment with variable impedance. A comprehensive overview of techniques for autonomous robotic ultrasound acquisitions is provided by Li et al. [4], and a broader review of robot-assisted ultrasound systems is given in [2].

All of the above systems rely on fixed-base manipulators and require the patient to be positioned within the arm's reachable workspace. In our own prior work [6], we demonstrated that a fixed collaborative robot arm can autonomously screen the abdominal aorta using impedance control and deep learning segmentation. The present paper extends this line of work to a fully mobile legged platform, removing the pre-positioning requirement and enabling operation in unstructured environments.

B. Whole-Body Control for Legged Manipulators

WBC has emerged as the dominant paradigm for coordinating locomotion and manipulation on legged robots. Khatib [13] introduced the operational space formulation that underpins most modern WBC approaches, enabling task-space control through the robot's full kinematic chain. Yoshikawa's manipulability measure [14] provides a scalar index of arm dexterity that has been widely used to avoid singular configurations.

For mobile manipulators, Bayle et al. [15] analysed the manipulability of wheeled platforms coupled with robot arms, showing that base motion can be used to maintain the arm in high-dexterity configurations. Haviland et al. [9] extended this to a holistic reactive formulation that simultaneously optimises base and arm velocities through a single QP, achieving smooth coordinated motion on wheeled platforms.

On legged robots, Bellicoso et al. [7] presented ALMA, demonstrating articulated locomotion and manipulation with a torque-controlled legged platform. Minniti et al. [16] proposed a whole-body Model Predictive Control (MPC) framework for dynamically stable mobile manipulation, and Sleiman et al. [8] unified locomotion and manipulation into a single MPC problem. These approaches have been validated primarily on ANYmal with structured targets at known locations. The Boston Dynamics Spot platform has been used for manipulation tasks with an external arm by Zimmermann et al. [17], who demonstrated dynamic grasping. However, none of the existing legged WBC systems have addressed perception-driven, multi-phase medical scanning tasks where the target must be detected and tracked throughout the mission.

C. Visual Perception for Patient Localisation

Accurate patient localisation from RGBD data is a prerequisite for autonomous mobile scanning. Skeleton-based approaches, which estimate 3D joint positions from monocular or depth-augmented images, have matured significantly with deep learning. YOLO-Pose [18] extended the YOLO detection framework to multi-person pose estimation using an object keypoint similarity loss, achieving real-time performance suitable for robot perception pipelines.

In our system, YOLO-based skeleton detection is combined with a KF for 3D torso tracking, providing robust localisation even under partial occlusion and during platform motion. The KF predicts torso position between detections and smooths noisy depth measurements from the Orbbec RGBD camera, enabling stable approach trajectories without requiring the robot to stop for reliable detection.

In summary, while individual components—robotic ultrasound, WBC, and visual patient tracking—have each received significant attention, no prior work has integrated them into a unified system capable of autonomously performing a medical scan on a legged platform navigating to an unknown patient location. The present paper addresses this gap.

III. PROBLEM FORMULATION

A. System Model

The system consists of a Boston Dynamics Spot quadruped and a Unitree Z1 serial manipulator arm rigidly mounted on the robot's back. Four reference frames are defined: the world frame $\mathcal{F}_W$, the Spot body frame $\mathcal{F}_B$ (origin at the geometric centre of the base), the arm base frame $\mathcal{F}_0$ (coincident with Z1 link 0, fixed relative to $\mathcal{F}_B$), and the end-effector frame $\mathcal{F}_{EE}$ (at the Z1 flange, where the ultrasound probe is attached).

1) *Spot base kinematics*: The Spot base is modelled as a non-holonomic vehicle constrained to move on a planar surface. Its configuration is described by $\xi = [x, y, \theta]^\top \in \mathbb{R}^2 \times \mathbb{S}^1$, where (x, y) is the position of $\mathcal{F}_B$ in $\mathcal{F}_W$ and θ is the yaw angle. The non-holonomic constraint (no lateral slip) restricts the admissible base velocities to

$$\dot{\xi} = \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} \cos \theta \\ \sin \theta \\ 0 \end{bmatrix} v_x + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \omega_z, \quad (1)$$

where $v_x \in \mathbb{R}$ is the forward linear velocity and $\omega_z \in \mathbb{R}$ is the yaw angular velocity, both expressed in $\mathcal{F}_B$.

2) *Z1 arm kinematics*: The Z1 arm has $n = 6$ revolute joints with configuration vector $\mathbf{q} = [q_1, \dots, q_6]^\top \in \mathbb{R}^6$ and joint velocity vector $\dot{\mathbf{q}} \in \mathbb{R}^6$. The forward kinematics map $\mathbf{T}_{EE}(\mathbf{q}) \in SE(3)$ gives the homogeneous transformation from $\mathcal{F}_0$ to $\mathcal{F}_{EE}$:

$$\mathbf{T}_{EE}(\mathbf{q}) = \prod_{i=1}^6 \mathbf{T}_{i-1,i}(q_i), \quad (2)$$

where $\mathbf{T}_{i-1,i}(q_i) \in SE(3)$ is the transformation between consecutive link frames. The arm geometric Jacobian $\mathbf{J}_{\text{arm}}(\mathbf{q}) \in \mathbb{R}^{6 \times 6}$ relates joint velocities to the spatial velocity of $\mathcal{F}_{EE}$ expressed in $\mathcal{F}_0$:

$$\mathbf{v}_{EE}^{\text{arm}} = \mathbf{J}_{\text{arm}}(\mathbf{q}) \dot{\mathbf{q}}, \quad (3)$$

where $\mathbf{v}_{EE}^{\text{arm}} = [\omega^\top, \dot{\mathbf{p}}^\top]^\top \in \mathbb{R}^6$ is the concatenation of angular velocity $\omega \in \mathbb{R}^3$ and linear velocity $\dot{\mathbf{p}} \in \mathbb{R}^3$ of the EE.

3) *Unified decision variable*: The combined control input is defined as

$$\mathbf{u} = \begin{bmatrix} \dot{\mathbf{q}} \\ v_x \\ \omega_z \end{bmatrix} \in \mathbb{R}^8, \quad (4)$$

subject to the velocity bounds

$$\mathbf{u}_{\min} \leq \mathbf{u} \leq \mathbf{u}_{\max}, \quad (5)$$

where $\mathbf{u}_{\min}, \mathbf{u}_{\max} \in \mathbb{R}^8$ encode joint velocity limits for the arm and maximum linear and angular speed for the base.

B. Task Definition

An autonomous FAST scan requires the EE to reach a sequence of N anatomical target poses. Let $\mathcal{T} = \{\mathbf{T}_i^* \in SE(3)\}_{i=1}^N$, with $N = 4$, denote the desired EE poses corresponding to the four FAST windows: subxiphoid, right upper quadrant, left upper quadrant, and suprapubic. Each target pose $\mathbf{T}_i^* = (\mathbf{R}_i^*, \mathbf{p}_i^*)$ comprises a desired EE position $\mathbf{p}_i^* \in \mathbb{R}^3$ and orientation $\mathbf{R}_i^* \in SO(3)$, both expressed in $\mathcal{F}_W$.

The target positions are not known a priori. During the approach phase, a patient position estimate $\hat{\mathbf{p}}(t) \in \mathbb{R}^3$, obtained from the YOLO+KF perception pipeline (Section V), is used to compute a dynamically updated intermediate goal for the WBC. Once the robot reaches the scanning distance, the FAST target poses $\mathcal{T}$ are determined from the surface geometry estimated by the EE-mounted RealSense camera.

C. Problem Statement

Given the system model defined in (1)–(4) and the target set $\mathcal{T}$, the autonomous FAST scanning problem is to find a control sequence $\mathbf{u}(t)$ that:

- 1) **Navigation**: drives the Spot base from its initial pose ξ_0 to a configuration from which all targets in $\mathcal{T}$ lie within the reachable workspace of the Z1 arm, using the online patient estimate $\hat{\mathbf{p}}(t)$;
- 2) **Probe placement**: brings the EE to each target pose $\mathbf{T}_i^* \in \mathcal{T}$ in sequence, satisfying

$$\|\mathbf{p}_{EE}(t_i) - \mathbf{p}_i^*\| \leq \varepsilon_p, \quad d(\mathbf{R}_{EE}(t_i), \mathbf{R}_i^*) \leq \varepsilon_r, \quad (6)$$

where $\mathbf{p}_{EE}$ and $\mathbf{R}_{EE}$ are the current EE position and orientation, $\varepsilon_p > 0$ is the admissible position error, $\varepsilon_r > 0$ is the admissible orientation error, and $d(\cdot, \cdot)$ denotes a rotation distance on $SO(3)$;

- 3) **Contact**: maintains the probe in compliant contact with the patient's skin during scanning, with normal contact force $f_c \in [f_{\min}, f_{\max}]$, where $f_{\min} > 0$ ensures acoustic coupling and $f_{\max}$ is the patient-safety limit;
- 4) **Input feasibility**: satisfies the velocity bounds in (5) at all times.

The following assumptions are adopted throughout this paper:

- A1** The patient remains stationary during the scanning phase.
- A2** Spot operates on flat terrain, so the non-holonomic model (1) is valid.
- A3** The rigid transform between $\mathcal{F}_B$ and $\mathcal{F}_0$ is known and constant (fixed mount).
- A4** A single patient is present in the scene.

IV. MOTION AND CONTACT CONTROL

The proposed system employs two complementary controllers that are active in different mission phases. During the *approach phase*, a WBC QP drives both the Spot base and the Z1 arm simultaneously to bring the EE close to the patient. During the *scanning phase*, a Cartesian impedance controller maintains compliant probe contact on the patient's skin. The handoff between the two controllers is managed by the mission FSM described in Section V. Fig. 1 shows the overall control loop.

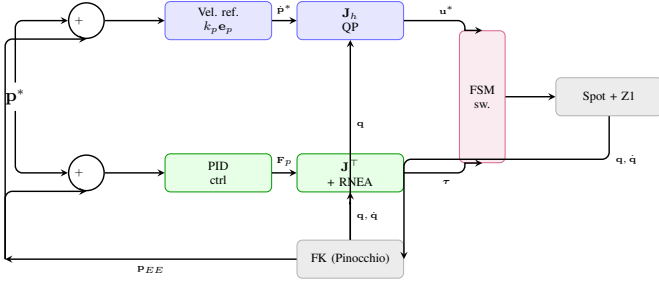


Fig. 1. Control loop architecture. During the approach phase the WBC QP generates joint and base velocity commands. During scanning the Cartesian impedance controller produces joint torques. The FSM switch selects the active controller. Both paths share the same forward kinematics feedback from Pinocchio.

A. Holistic Jacobian

To enable simultaneous base and arm control, we construct a holistic Jacobian that maps the unified control input $\mathbf{u}$ (defined in (4)) to the spatial velocity of the EE expressed in the world frame $\mathcal{F}_W$.

1) *Base contribution*: The Spot base moves as a non-holonomic vehicle. The linear velocity of the EE induced by base motion depends on both v_x and ω_z and on the position of the EE relative to the body frame origin. Let $\mathbf{p}_{EE}^B \in \mathbb{R}^3$ be the EE position expressed in $\mathcal{F}_B$, and let $\mathbf{R}_{WB} \in SO(3)$ be the rotation matrix from $\mathcal{F}_B$ to $\mathcal{F}_W$. The contribution of base velocities $[v_x, \omega_z]^T$ to the spatial EE velocity in $\mathcal{F}_W$ is captured by the base Jacobian $\mathbf{J}_{base} \in \mathbb{R}^{6 \times 2}$:

$$\mathbf{J}_{base} = \mathbf{R}_{WB} \begin{bmatrix} \mathbf{e}_x & \mathbf{0}_{3 \times 2} \\ \omega_z \times \mathbf{p}_{EE}^B \end{bmatrix}, \quad (7)$$

where $\mathbf{e}_x = [1, 0, 0]^T$ is the forward direction in $\mathcal{F}_B$ and $\omega_z = [0, 0, 1]^T$ is the yaw axis. The upper three rows (angular part) are zero because the non-holonomic base does not contribute to the EE angular velocity independently of the arm joints.

2) *Holistic Jacobian assembly*: The arm Jacobian $\mathbf{J}_{arm}(\mathbf{q}) \in \mathbb{R}^{6 \times 6}$ (defined in (3)) is rotated into $\mathcal{F}_W$ using the current body orientation $\mathbf{R}_{WB}$. The holistic Jacobian is then assembled by horizontal concatenation:

$$\mathbf{J}_h(\mathbf{q}, \xi) = [\mathbf{R}_{WB} \mathbf{J}_{arm}(\mathbf{q}) \mid \mathbf{J}_{base}(\mathbf{q}, \xi)] \in \mathbb{R}^{6 \times 8}, \quad (8)$$

so that the EE spatial velocity satisfies $\mathbf{v}_{EE} = \mathbf{J}_h \mathbf{u}$.

B. WBC Quadratic Program

1) *Manipulability measure*: The Yoshikawa manipulability index [14] is used to quantify the arm's distance from kinematic singularities:

$$m(\mathbf{q}) = \sqrt{\det(\mathbf{J}_{arm}(\mathbf{q}) \mathbf{J}_{arm}^T(\mathbf{q}))}. \quad (9)$$

When m is large the arm is well-conditioned; when $m \rightarrow 0$ the arm approaches a singularity.

2) *Weighted cost function*: The WBC QP seeks the control input $\mathbf{u}$ that best tracks a desired EE velocity $\mathbf{v}_{des} \in \mathbb{R}^6$ while penalising joint and base velocities through a manipulability-aware weight matrix. The optimisation problem is:

$$\mathbf{u}^* = \arg \min_{\mathbf{u}} \frac{1}{2} \mathbf{u}^T \mathbf{W} \mathbf{u} \quad \text{s.t.} \quad \mathbf{J}_h \mathbf{u} \approx \mathbf{v}_{des}, \quad (10)$$

where $\mathbf{W} = \text{diag}(w_1, \dots, w_8) \in \mathbb{R}^{8 \times 8}$ is a diagonal weight matrix with entries:

$$w_i = \begin{cases} \frac{\lambda_{arm}}{m(\mathbf{q}) + \varepsilon} & i = 1, \dots, 6 \quad (\text{arm joints}) \\ \lambda_{base} & i = 7, 8 \quad (\text{base: } v_x, \omega_z) \end{cases} \quad (11)$$

Here $\lambda_{arm}, \lambda_{base} > 0$ are tuning gains and $\varepsilon > 0$ is a small regularisation constant that prevents division by zero. When m is high (arm well-conditioned), arm weights decrease, making arm motion less costly and preferred over base motion. As $m \rightarrow 0$ (near singularity), arm weights increase, automatically transferring effort to the base.

3) *Closed-form solution*: The constrained QP (10) admits a closed-form solution via the weighted damped pseudo-inverse:

$$\mathbf{u}^* = \mathbf{W}^{-1} \mathbf{J}_h^T (\mathbf{J}_h \mathbf{W}^{-1} \mathbf{J}_h^T + \mu \mathbf{I}_6)^{-1} \mathbf{v}_{des}, \quad (12)$$

where $\mu > 0$ is a damping factor that ensures numerical stability near singularities. The solution is projected back to feasible commands by clipping each component of $\mathbf{u}^*$ to the bounds in (5).

4) *Desired EE velocity*: The desired EE velocity is computed from the position error between the current EE position $\mathbf{p}_{EE} \in \mathbb{R}^3$ and the target position $\mathbf{p}^* \in \mathbb{R}^3$ provided by the perception pipeline:

$$\mathbf{v}_{des} = \begin{bmatrix} \mathbf{0}_3 \\ k_p (\mathbf{p}^* - \mathbf{p}_{EE}) \end{bmatrix}, \quad (13)$$

where $k_p > 0$ is a proportional gain with units of s^{-1} . The angular component is set to zero as orientation tracking during approach is not required; probe orientation is managed by the impedance controller during the scanning phase.

C. Cartesian Impedance Control

Once the FSM transitions to the scanning phase, the WBC QP is deactivated and a Cartesian impedance controller takes over the Z1 arm. The controller implements a hybrid strategy: Cartesian Proportional-Integral-Derivative (PID) control for translational motion and joint-space Proportional-Derivative (PD) control for wrist orientation.

1) *Cartesian position control*: Let $\mathbf{p}_{EE}(t) \in \mathbb{R}^3$ be the current EE position and $\mathbf{p}_d(t) \in \mathbb{R}^3$ the desired position, updated at 500 Hz from the RealSense surface estimate. The translational position error is:

$$\mathbf{e}_p(t) = \mathbf{p}_d(t) - \mathbf{p}_{EE}(t). \quad (14)$$

The Cartesian control force is computed as:

$$\mathbf{F}_p = \mathbf{K}_p^{\text{eff}} \mathbf{e}_p - \mathbf{K}_d^{\text{eff}} \dot{\mathbf{p}}_{EE} + \mathbf{K}_i \int \mathbf{e}_p dt, \quad (15)$$

where $\dot{\mathbf{p}}_{EE} = \mathbf{J}_{\text{arm}} \dot{\mathbf{q}}$ is the EE translational velocity, and $\mathbf{K}_p^{\text{eff}}, \mathbf{K}_d^{\text{eff}} \in \mathbb{R}^{3 \times 3}$ are effective diagonal gain matrices that adapt to the current arm extension:

$$\mathbf{K}_p^{\text{eff}} = (1 - 0.65\rho) \mathbf{K}_p, \quad \mathbf{K}_d^{\text{eff}} = (1 + \rho) \mathbf{K}_d, \quad (16)$$

where $\rho = \text{clip}(\|\mathbf{p}_{EE}^{xy}\|/r_{\text{max}}, 0, 1) \in [0, 1]$ is the normalised planar reach, $\mathbf{p}_{EE}^{xy}$ is the horizontal EE position in $\mathcal{F}_0$, and $r_{\text{max}} = 0.6$ m is the arm's nominal planar reach. This adaptive scaling reduces stiffness and increases damping as the arm extends, improving stability at the workspace boundary. The nominal gains are $\mathbf{K}_p = \text{diag}(250, 250, 300) \text{ N m}^{-1}$ and $\mathbf{K}_d = \text{diag}(15, 15, 15) \text{ N s m}^{-1}$.

2) *Surface-referenced desired position*: When the RealSense surface estimate is available, the desired position is computed relative to the detected patient surface. Let $\mathbf{p}_s \in \mathbb{R}^3$ be the surface centroid and $\mathbf{n}_s \in \mathbb{S}^2$ the outward surface normal estimated by the RealSense pipeline. The desired EE position is:

$$\mathbf{p}_d = \mathbf{p}_s + \delta_n \mathbf{n}_s, \quad (17)$$

where $\delta_n = -5$ mm is a small offset along the inward normal that ensures light contact without excessive force.

3) *Joint torque computation*: The Cartesian force $\mathbf{F}_p$ is projected into joint torques via the arm Jacobian transpose, and combined with joint-space wrist control and dynamic compensation:

$$\boldsymbol{\tau} = \mathbf{J}_{\text{arm}}^\top \tilde{\mathbf{F}}_p + \boldsymbol{\tau}_{\text{wrist}} + \boldsymbol{\tau}_{\text{comp}}, \quad (18)$$

where $\tilde{\mathbf{F}}_p = [\mathbf{F}_p^\top, \mathbf{0}_3^\top]^\top \in \mathbb{R}^6$ pads the translational force with zero moments (rotational Cartesian control is disabled), and each term is:

- $\boldsymbol{\tau}_{\text{wrist}} \in \mathbb{R}^6$: joint-space PD control applied to wrist joints q_4, q_5, q_6 to maintain probe perpendicularity to the surface:

$$\tau_{\text{wrist},i} = K_{p,w}(q_{d,i} - q_i) - K_{d,w}\dot{q}_i, \quad i = 4, 5, 6, \quad (19)$$

with $K_{p,w} = 40 \text{ N m rad}^{-1}$ and $K_{d,w} = 4 \text{ N m s rad}^{-1}$;

- $\boldsymbol{\tau}_{\text{comp}} \in \mathbb{R}^6$: gravity and Coriolis compensation computed via the Recursive Newton-Euler Algorithm (RNEA) implemented in Pinocchio [19]:

$$\boldsymbol{\tau}_{\text{comp}} = \boldsymbol{\tau}_{\text{gravity}}(\mathbf{q}) + \boldsymbol{\tau}_{\text{Coriolis}}(\mathbf{q}, \dot{\mathbf{q}}). \quad (20)$$

The total torque is saturated to the hardware limit $\tau_{\text{max}} = 70 \text{ N m}$ before being sent to the joint controllers:

$$\boldsymbol{\tau}_{\text{cmd}} = \text{clip}(\boldsymbol{\tau}, -\tau_{\text{max}} \mathbf{1}, \tau_{\text{max}} \mathbf{1}). \quad (21)$$

V. SYSTEM ARCHITECTURE

Fig. 2 gives an overview of the full TERESA system. The architecture is organised into four layers: sensors, perception, mission management, and control. All components communicate over Robot Operating System (ROS) 2 with Data Distribution Service (DDS), distributed across two compute units: a NVIDIA Jetson (perception and arm control) and the Spot onboard computer SpotCore (base control and locomotion).

A. Hardware Platform

The TERESA platform consists of three main hardware components. The **Boston Dynamics Spot** quadruped provides mobile navigation on unstructured terrain. It is commanded via its onboard ROS 2 driver (`spot_ros2`) through high-level velocity commands $[v_x, \omega_z]$ on the `/cmd_vel` topic. The **Unitree Z1** is a six-DOF serial manipulator with a payload of 2 kg, mounted rigidly on Spot's back. It is controlled at the torque level through a Joint Trajectory Controller (JTC) interface running on the Jetson. The **ultrasound probe** is attached to the Z1 flange (EE frame $\mathcal{F}_{EE}$) and serves as the sensing element for the FAST scan.

B. Sensor Suite

Two depth cameras cover complementary fields of view:

- **Orbbec RGB-D**: mounted on the Spot body, provides a wide field of view for patient detection at distances up to 3 m. It feeds the YOLO+KF perception pipeline during the approach phase.
- **Intel RealSense D435**: mounted at the Z1 EE, provides close-range depth and colour for surface normal estimation. It is active during the scanning phase once the probe is within 30 cm of the patient.

The static transform between the Spot body frame $\mathcal{F}_B$ and the arm base frame $\mathcal{F}_0$ is a calibrated rigid offset published as a static Transform (TF) at system startup.

C. Perception Pipeline

1) *Wide-field patient localisation (Orbbec)*: During the approach phase, YOLO11-pose [18] runs on the Orbbec RGBD stream at 30 Hz and detects the skeleton of the patient. The torso position is computed as the centroid of four Common Objects in Context (COCO) keypoints—left shoulder, right shoulder, left hip, right hip—by back-projecting their image coordinates through the calibrated depth map. The resulting 3D measurement feeds the Kalman3D filter (state: position + velocity, $\mathbf{x}_{KF} \in \mathbb{R}^6$), which predicts the torso position at every frame and updates only when a valid detection is available. The filtered estimate $\hat{\mathbf{p}}(t)$ is published to the WBC as the approach target.

2) *Close-range surface estimation (RealSense)*: Once the FSM transitions to the HANDOFF state, the RealSense becomes the primary sensor. A surface estimation node fits a plane to the depth point cloud in a region of interest around the detected torso, and publishes the surface centroid $\mathbf{p}_s$ and outward normal $\mathbf{n}_s$ as a `PoseStamped` message. These are used by the impedance controller to compute the desired probe position via (17).

D. Mission FSM

The mission is orchestrated by a six-state FSM implemented in the `wbc_coordinator` node. Fig. 3 shows the state diagram.

- **IDLE**: system inactive, waiting for a valid patient detection from the Orbbec pipeline.

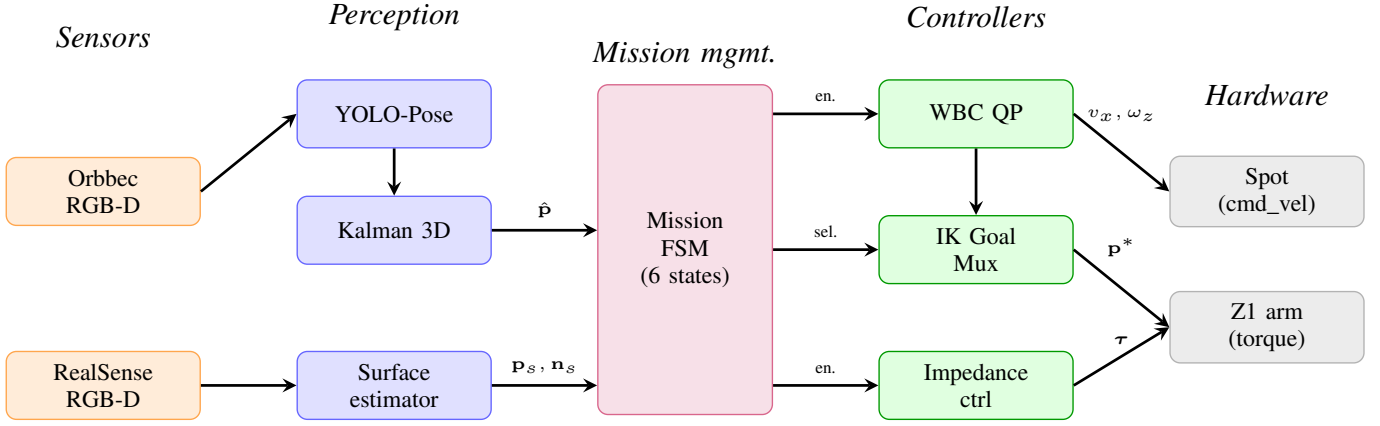


Fig. 2. Block diagram of the TERESA system. Arrows indicate the main data flows. The FSM acts as the central orchestrator, enabling and disabling the WBC QP and impedance controller based on mission phase.

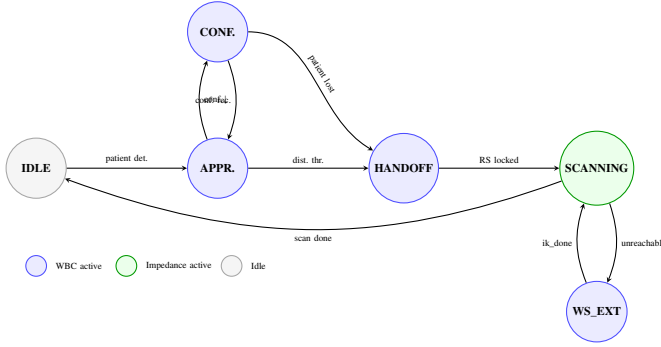


Fig. 3. Mission FSM. Blue states have the WBC QP active; the green state uses the impedance controller. APPR. = APPROACHING, CONF. = CONFIRMING, WS_EXT = WS_EXTENSION.

- **APPROACHING:** WBC QP active. Spot navigates toward the patient while the Z1 arm pre-positions the probe in the direction of $\hat{p}(t)$. The QP target is updated every 1.5 s to respect the minimum trajectory time of the arm JTC.
- **CONFIRMING:** Orbbec detection confidence drops below threshold. The arm holds its last estimate; the WBC remains active but the goal is frozen. If confidence recovers, the FSM returns to APPROACHING; if the patient is completely lost, it transitions to HANDOFF.
- **HANDOFF:** Orbbec loses the patient entirely. Spot halts. Control is transferred to the RealSense pipeline, which must lock onto the torso surface before the FSM can advance.
- **SCANNING:** WBC QP dormant. The impedance controller drives the probe through the FAST scan sequence using the surface frame from the RealSense. If a target is reported unreachable, the FSM triggers a workspace extension.
- **WS_EXTENSION:** the WBC QP reactivates to execute a micro-step that repositions Spot so that the target re-enters the Z1 workspace. Once the arm Inverse Kinemat-

ics (IK) confirms success (`/ik_done`), the FSM returns to SCANNING.

E. IK Goal Multiplexer

The Z1 arm IK-to-JTC node receives a single `/ik_goal_pose` topic. Since both the WBC QP and the `z1_fsm` scan controller publish arm targets, an `ik_goal_mux` node arbitrates between them based on a binary enable signal (`/wbc/enable`) from the FSM: when WBC is active its target takes priority; otherwise the `z1_fsm` target is forwarded. This design preserves the existing arm control stack unchanged.

VI. EXPERIMENTAL SETUP

[TODO: Write experimental setup]

VII. RESULTS AND DISCUSSION

[TODO: Write results and discussion]

VIII. CONCLUSION

[TODO: Write conclusion]

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